

Program: Diploma in Automation and Robotics	
Course Code:	Course Title: Electrical Circuits and Machines
Semester: 3	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To provide a basic knowledge about the fundamental concepts and circuit laws for the analysis of basic DC and AC circuits
- To acquire the basic knowledge of construction, working and operation of DC machine, transformer and induction motor.
- Able to select a particular transformer/DC motor/Induction motor depending on the application

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic principles and theorems of Engineering Mathematics		Mathematics I &II	1 & 2
Fundamentals of electrical engineering	2031	Fundamentals of Electrical and Electronics Engineering	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Apply fundamental concepts and circuit laws to solve simple DC electric circuits and AC circuits.	15	Applying
CO2	Realize the concept of resonance and power measurement by two wattmeter method	13	Applying
CO3	Describe the principle of operation of DC generators and motors and select appropriate motor types for different applications.	15	Applying
CO4	Realize the principle of operation and requirement of AC machines and transformers in transmission and distribution of electric power and other applications	15	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1: Apply fundamental concepts and circuit laws to solve simple DC electric circuits and AC circuits			
M1.01	Explain Kirchhoff's law and solve simple problems	3	Applying
M1.02	Solve problems in mesh current and nodal voltage method	3	Applying
M1.03	Solve problems in Super position, Thevenin's, Norton 's and maximum power transfer theorems	3	Applying
M1.04	Solve problems in RLC series and parallel circuits	4	Applying
M1.05	Concept of Harmonics	2	Understanding
DC NETWORK THEOREMS Kirchhoff 's laws – Mesh Current Method, Nodal Voltage Method, Super position, Thevenin 's, Norton 's and maximum power transfer theorems – Statement and explanations – Simple problems AC CIRCUITS Average value – RMS value (effective value) – form factor – crest factor – Rectangular and polar forms. AC Through pure resistor, inductor and Capacitor – Concept of impedance – vector diagram. Capacitors in series and parallel – energy stored in a capacitor– derivation – simple problems. Power in AC circuits – power factor– RL, RC and RLC series and parallel circuits – simple problems. Introduction of Harmonics - Effects of Harmonics			
CO2	Realize the concept of resonance and power measurement by two wattmeter method		
M2.01	Explain the concept of resonance, types of resonance and applications of resonance	4	Understanding
M2.02	Simple problems related to resonance	2	Applying
M2.03	Concept of three phase power, line voltage, phase voltage and current in star and delta connected	5	Applying

	circuit and simple problems		
M2.04	Measurement of three-phase power by two wattmeter method and simple problems	2	Applying
	Series Test – I	1	
Resonance and 3ϕ AC circuits Resonance – condition for resonance – series and parallel resonance – resonance curve – effect of resistance on resonance curve – selectivity – Q factor and bandwidth – applications of resonance – simple problems in resonance. Concept of 3 ϕ supply – line and phase voltage and current in star and delta connected circuits – three phase power – Measurement of three phase power by two-watt meter method – simple problems – advantages of three phase over single-phase system			
CO3	Describe the principle of operation of DC generators and motors and select appropriate motor types for different applications.		
M3.01	Acquire knowledge about constructional details of DC machines	2	Understanding
M3.02	Describe the principle of operation, types and performance characteristics of DC generators	3	Understanding
M3.03	Describe the principle of operation, types and characteristics of DC motors	3	Understanding
M3.04	Understand the working of starters	3	Understanding
M3.05	Understand the various speed control techniques of DC motor	4	Understanding
DC machines – Types – constructional details of DC machines – DC generators – principle – types – emf equation – characteristics of shunt, series and compound generators. DC motor – types – motor action – back emf – torque speed characteristics – starting of motors using 3- and 4-point starters – speed control of DC motor-applications.			
CO4: Realize the principle of operation and requirement of AC machines and transformers in transmission and distribution of electric power and other applications			
M4.01	Understand the construction, operating principle and characteristics of three phase alternator and three phase induction motor.	3	Understanding
M4.02	methods of starting of 3 ϕ induction motor	2	Understanding
M4.03	Explain the working, principle of operation and starting methods of single-phase induction motor	3	Understanding
M4.04	Understand the construction, operating principle and characteristics of single-phase transformer	2	Understanding
M4.05	Simple problems related to SC and OC test, efficiency and regulation	3	Applying
M4.06	Explain the construction and working of autotransformer	2	Understanding
	Series Test – II	1	
AC machines and Transformers AC machines – 3 ϕ alternator – construction and working – relation between speed and frequency. 3 ϕ Induction motor – construction – types – principle of operation – methods of starting of 3 ϕ induction motor – slip. Single phase induction motor – principle of operation – split phase and capacitor start motors –			

Applications – principle of operation - Transformer – Ideal transformer – principle of working – constructional details – emf equation – turns ratio – core loss – copper loss – efficiency – regulation – SC and OC tests – simple problems. Transformer on No load – Transformer on load – condition for maximum efficiency All-day efficiency (simple problems). Auto transformer – construction and working – applications.

Text / Reference

T/R	Book Title/Author
T1	Bhattacharya, S. K., Basic Electrical & Electronics Engineering, Pearson
T2	Nagoor kani , Circuit Theory ,RBA Publications
T3	Theraja. B.L., A text book of Electrical Technology, Vol. I & III, S.Chand & Co.
R3	Arumugam & Prem kumar, Circuit Theory, Khanna Publishers
R4	Louis M.M. ,Elements of Electrical Engineering , Khanna Publishers
R5	Gupta M.L., S.K.Kataria & Sons, Elementary of Electrical Engineering

Online Resources

SI No	Website Link
1	https://www.electrical4u.com/
2	https://nptel.ac.in/courses/108/108/108108076/